Zero inflated models

Data: College drinking

Survey data from 77 college students on a dry campus (i.e., alcohol is prohibited) in the US. Survey asks students "How many alcoholic drinks did you consume last weekend?"

- `drinks`: the number of drinks the student reports consuming
- `sex`: an indicator for whether the student identifies as male
- `OffCampus`: an indicator for whether the student lives off campus
- `FirstYear`: an indicator for whether the student is a first-year student

> Our goal: model the number of drinks students report consuming.

Class activity: The fitted model

$$P(Y_i = y) = \begin{cases} e^{-\lambda_i}(1-\alpha_i) + \alpha_i & y = 0 \\ \dfrac{e^{-\lambda_i}\lambda_i^y}{y!}(1-\alpha_i) & y > 0 \end{cases}$$

$$\log\left(\frac{\widehat{\alpha}_i}{1-\widehat{\alpha}_i}\right) = -0.60 + 1.14 FirstYear_i$$

$$\log(\widehat{\lambda}_i) = 0.75 + 0.42\, OffCampus_i + 1.02\, Male_i$$

> What is the estimated probability that a first year student never drinks?

The fitted model

$$P(Y_i = y) = \begin{cases} e^{-\lambda_i}(1-\alpha_i) + \alpha_i & y = 0 \\ \dfrac{e^{-\lambda_i}\lambda_i^y}{y!}(1-\alpha_i) & y > 0 \end{cases}$$

$$\log\left(\frac{\widehat{\alpha}_i}{1-\widehat{\alpha}_i}\right) = -0.60 + 1.14 FirstYear_i$$

$$\log(\widehat{\lambda}_i) = 0.75 + 0.42\, OffCampus_i + 1.02\, Male_i$$

> What is the estimated average number of drinks for a male student who lives off campus and sometimes drinks?

The fitted model

$$P(Y_i = y) = \begin{cases} e^{-\lambda_i}(1-\alpha_i) + \alpha_i & y = 0 \\ \dfrac{e^{-\lambda_i}\lambda_i^y}{y!}(1-\alpha_i) & y > 0 \end{cases}$$

$$\log\left(\frac{\widehat{\alpha}_i}{1-\widehat{\alpha}_i}\right) = -0.60 + 1.14 FirstYear_i$$

$$\log(\widehat{\lambda}_i) = 0.75 + 0.42\ OffCampus_i + 1.02\ Male_i$$

> What is the estimated probability that a male first year student who lives off campus had at least one drink last weekend?

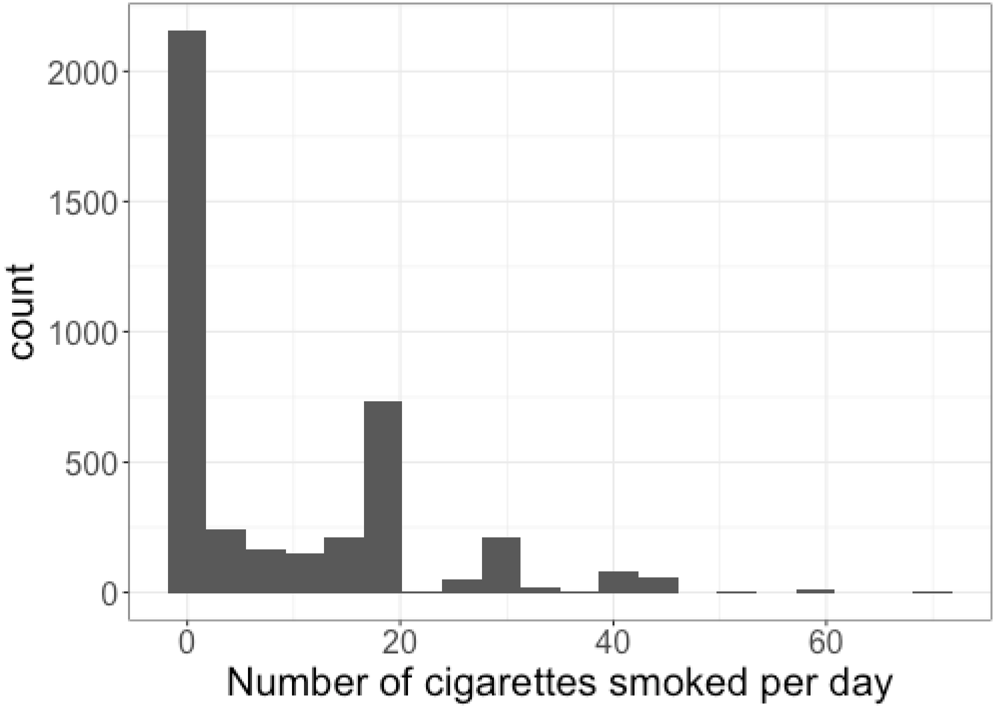
Data: Framingham heart study

Data collected on residents of Framingham, MA over a long period of time, to study variables related to heart health. We will work with a subset of the data, containing

- `cigsPerDay`: The number of cigarettes smoked per day during the study period.
- `education`: 1 = High School, 2 = Some College, 3 = College Degree, 4 = Advanced Degree.
- `male`: 1 = Male, 0 = Female.
- `age`: The age of the individual in years.
- `diabetes`: 1 if the individual has diabetes, 0 otherwise.

> Why might we see zero inflation in the number of cigarettes smoked per day?

EDA: number of cigarettes smoked



Class activity

Work on the class activity (handout).

Class activity

$$P(Y_i = y) = \begin{cases} e^{-\lambda_i}(1-\alpha_i) + \alpha_i & y = 0 \\ \dfrac{e^{-\lambda_i}\lambda_i^y}{y!}(1-\alpha_i) & y > 0 \end{cases}$$

$$\log\left(\frac{\widehat{\alpha}_i}{1-\widehat{\alpha}_i}\right) = -2.51 + 0.051 Age_i$$

$$\log(\widehat{\lambda}_i) = 3.21 - 0.044 EducationSome_i - 0.082 EducationCollege_i - 0.0062 EducationAdv_i - 0.024 Diabetes_i - 0.0056 Age_i$$

> How do we interpret the coefficient -0.024 in the fitted model?

Class activity

$$\log\left(\frac{\widehat{\alpha}_i}{1-\widehat{\alpha}_i}\right) = -2.51 + 0.051 Age_i$$

$$\log(\widehat{\lambda}_i) = 3.21 - 0.044 EducationSome_i - 0.082 EducationCollege_i - 0.0062 EducationAdv_i - 0.024 Diabetes_i - 0.0056 Age_i$$

> What is the expected number of cigarettes smoked per day, for a 50 year old smoker with diabetes and some college education?

Making predictions

$$\log\left(\frac{\widehat{\alpha}_i}{1-\widehat{\alpha}_i}\right) = -2.51 + 0.051 Age_i$$

$$\log(\widehat{\lambda}_i) = 3.21 - 0.044 EducationSome_i - 0.082 EducationCollege_i - 0.0062 EducationAdv_i - 0.024 Diabetes_i - 0.0056 Age_i$$

> What if I want the expected number of cigarettes smoked per day for a 50 year old with diabetes and some college education, but *without* knowing whether they are a smoker?

A new question

$$P(Y_i = y) = \begin{cases} e^{-\lambda_i}(1-\alpha_i) + \alpha_i & y = 0 \\ \dfrac{e^{-\lambda_i}\lambda_i^y}{y!}(1-\alpha_i) & y > 0 \end{cases}$$

$$\log\left(\frac{\alpha_i}{1-\alpha_i}\right) = \gamma_0 + \gamma_1 Age_i$$

$$\log(\lambda_i) = \beta_0 + \beta_1 EducationSome_i + \beta_2 EducationCollege_i + \beta_3 EducationAdv_i + \beta_4 Diabetes_i + \beta_5 Age_i$$

New research question: for smokers, does the number of cigarettes smoked per day depend on age?

How would we answer this research question?

Fitting the model in R

```
m2 <- zeroinfl(cigsPerDay ~ education +
                 diabetes + age | age,
               data = heart_data)
summary(m2)
```

```
Count model coefficients (poisson with log link):
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  3.2063376  0.0342290  93.673  < 2e-16 ***
education2  -0.0441182  0.0124809  -3.535 0.000408 ***
education3  -0.0820372  0.0158604  -5.172 2.31e-07 ***
education4  -0.0062453  0.0171640  -0.364 0.715964
diabetes    -0.0241459  0.0386337  -0.625 0.531974
age         -0.0056182  0.0006738  -8.338  < 2e-16 ***

Zero-inflation model coefficients (binomial with logit link):
             Estimate Std. Error z value Pr(>|z|)
(Intercept) -2.506681   0.191483  -13.09   <2e-16 ***
age          0.051329   0.003821   13.43   <2e-16 ***
```

z-test

```
m2 <- zeroinfl(cigsPerDay ~ education +
                 diabetes + age | age,
               data = heart_data)
summary(m2)
```

```
Count model coefficients (poisson with log link):
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  3.2063376  0.0342290  93.673  < 2e-16 ***
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age         -0.0056182  0.0006738  -8.338  < 2e-16 ***
```

Likelihood ratio test

Likelihood ratio test

```
m2 <- zeroinfl(cigsPerDay ~ education +
                 diabetes + age | age,
               data = heart_data)
m2$loglik
```

```
## [1] -14023.42
```

```
m1 <- zeroinfl(cigsPerDay ~ education +
                 diabetes | age,
               data = heart_data)
m1$loglik
```

```
## [1] -14058.41
```

Likelihood ratio test

```
m1 <- zeroinfl(cigsPerDay ~ education +
                 diabetes | age,
               data = heart_data)

m2 <- zeroinfl(cigsPerDay ~ education +
                 diabetes + age | age,
               data = heart_data)

2*(m2$loglik - m1$loglik)
```

```
## [1] 69.97593
```

```
pchisq(69.976, df=1, lower.tail=F)
```

```
## [1] 6.003042e-17
```

Class activity

Work on the class activity. At the end of class, render your work as an HTML and submit on Canvas.